## 1.3.3 The Frobenius norm ¶

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**Definition 1.3.3.1. The Frobenius norm.** The Frobenius norm $\| \cdot \|_F : \mathbb{C}^{m \times n} \rightarrow \mathbb{R}$ is defined for $A \in \mathbb{C}^{m \times n}$ by

$$\|A\|_F = \sqrt{\sum_{i=0}^{m-1} \sum_{j=0}^{n-1} |\alpha_{i,j}|^2} = \sqrt{\begin{array}{ccccccc} |\alpha_{0,0}|^2 & + & \cdots & + & |\alpha_{0,n-1}|^2 & + \\ \vdots & \vdots & & \vdots & \vdots & \vdots \\ |\alpha_{m-1,0}|^2 & + & \cdots & + & |\alpha_{m-1,n-1}|^2. \end{array}}$$

One can think of the Frobenius norm as taking the columns of the matrix, stacking them on top of each other to create a vector of size $m \times n$, and then taking the vector 2-norm of the result.

**Homework 1.3.3.1.** Partition $m \times n$ matrix $A$ by columns:

$$A = \left( \begin{array}{c|c|c} a_0 & \cdots & a_{n-1} \end{array} \right).$$

Show that

$$\|A\|_F^2 = \sum_{j=0}^{n-1} \|a_j\|_2^2.$$

▼ Solution

$$
\begin{array}{l}
\|A\|_F \\
\quad = \quad \langle \text{ definition } \rangle \\
\sqrt{\sum_{i=0}^{m-1}\sum_{j=0}^{n-1}|\alpha_{i,j}|^2} \\
\quad = \quad \langle \text{ commutativity of addition } \rangle \\
\sqrt{\sum_{j=0}^{n-1}\sum_{i=0}^{m-1}|\alpha_{i,j}|^2} \\
\quad = \quad \langle \text{ definition of vector 2-norm } \rangle \\
\sqrt{\sum_{j=0}^{n-1}\|a_j\|_2^2}
\end{array}
$$

**Homework 1.3.3.2.** Prove that the Frobenius norm is a norm.

▼ Solution

Establishing that this function is positive definite and homogeneous is straight forward. To show that the triangle inequality holds it helps to realize that if $A = \left( \begin{array}{c|c|c|c} a_0 & a_1 & \cdots & a_{n-1} \end{array} \right)$ then

$$
\begin{array}{l}
\|A\|_F \\
\quad = \quad \langle \text{ definition } \rangle \\
\sqrt{\sum_{i=0}^{m-1}\sum_{j=0}^{n-1}|\alpha_{i,j}|^2} \\
\quad = \quad \langle \text{ commutativity of addition } \rangle \\
\sqrt{\sum_{j=0}^{n-1}\sum_{i=0}^{m-1}|\alpha_{i,j}|^2} \\
\quad = \quad \langle \text{ definition of vector 2-norm } \rangle \\
\sqrt{\sum_{j=0}^{n-1}\|a_j\|_2^2} \\
\quad = \quad \langle \text{ definition of vector 2-norm } \rangle \\
\sqrt{\left\| \begin{pmatrix} a_0 \\ a_1 \\ \vdots \\ a_{n-1} \end{pmatrix} \right\|_2^2}.
\end{array}
$$

In other words, it equals the vector 2-norm of the vector that is created by stacking the columns of $A$ on top of each other. One can then exploit the fact that the vector 2-norm obeys the triangle inequality.

**Homework 1.3.3.3.** Partition $m \times n$ matrix $A$ by rows:

$$A = \left( \begin{array}{c} \tilde{a}_0^T \\ \hline \vdots \\ \hline \tilde{a}_{m-1}^T \end{array} \right).$$

Show that

$$\| A \|_F^2 = \sum_{i=0}^{m-1} \| \tilde{a}_i \|_2^2,$$

where $\tilde{a}_i = {\tilde{a}_i^T}^T$.

▼ Solution

$\| A \|_F$

$=$ $<$ definition $>$

$\sqrt{\sum_{i=0}^{m-1} \sum_{j=0}^{n-1} |\alpha_{i,j}|^2}$

$=$ $<$ definition of vector 2-norm $>$

$\sqrt{\sum_{i=0}^{m-1} \| \tilde{a}_i \|_2^2}.$

Let us review the definition of the transpose of a matrix (which we have already used when defining the dot product of two real-valued vectors and when identifying a row in a matrix):

**Definition 1.3.3.2. Transpose.** If $A \in \mathbb{C}^{m \times n}$ and

$$A = \left( \begin{array}{c|c|c|c} \alpha_{0,0} & \alpha_{0,1} & \cdots & \alpha_{0,n-1} \\ \hline \alpha_{1,0} & \alpha_{1,1} & \cdots & \alpha_{1,n-1} \\ \hline \vdots & \vdots & & \vdots \\ \vdots & & & \\ \hline \alpha_{m-1,0} & \alpha_{m-1,1} & \cdots & \alpha_{m-1,n-1} \end{array} \right)$$

then its ***transpose*** is defined by

$$A^T = \left( \begin{array}{c|c|c|c} \alpha_{0,0} & \alpha_{1,0} & \cdots & \alpha_{m-1,0} \\ \hline \alpha_{0,1} & \alpha_{1,1} & \cdots & \alpha_{m-1,1} \\ \hline \vdots & \vdots & & \vdots \\ \vdots & & & \\ \hline \alpha_{0,n-1} & \alpha_{1,n-1} & \cdots & \alpha_{m-1,n-1} \end{array} \right).$$

For complex-valued matrices, it is important to also define the ***Hermitian transpose*** of a matrix:

**Definition 1.3.3.3. Hermitian transpose.** If $A \in \mathbb{C}^{m \times n}$ and

$$A = \left( \begin{array}{c|c|c|c} \alpha_{0,0} & \alpha_{0,1} & \cdots & \alpha_{0,n-1} \\ \hline \alpha_{1,0} & \alpha_{1,1} & \cdots & \alpha_{1,n-1} \\ \hline \vdots & \vdots & & \vdots \\ \vdots & & & \\ \hline \alpha_{m-1,0} & \alpha_{m-1,1} & \cdots & \alpha_{m-1,n-1} \end{array} \right)$$

then its ***Hermitian transpose*** is defined by

$$A^H = \overline{A}^T \left( \begin{array}{c|c|c|c} \overline{\alpha}_{0,0} & \overline{\alpha}_{1,0} & \cdots & \overline{\alpha}_{m-1,0} \\ \hline \overline{\alpha}_{0,1} & \overline{\alpha}_{1,1} & \cdots & \overline{\alpha}_{m-1,1} \\ \hline \vdots & \vdots & & \vdots \\ \vdots & & & \\ \hline \overline{\alpha}_{0,n-1} & \overline{\alpha}_{1,n-1} & \cdots & \overline{\alpha}_{m-1,n-1} \end{array} \right),$$

where $\overline{A}$ denotes the ***conjugate of a matrix***, in which each element of the matrix is conjugated.

We note that

- $\overline{A}^T = \overline{A^T}$.
- If $A \in \mathbb{R}^{m \times n}$, then $A^H = A^T$.
- If $x \in \mathbb{C}^m$, then $x^H$ is defined consistent with how we have used it before.
- If $\alpha \in \mathbb{C}$, then $\alpha^H = \overline{\alpha}$.

  (If you view the scalar as a matrix and then Hermitian transpose it, you get the matrix with as only element $\overline{\alpha}$.)

*Don't Panic!*. While working with complex-valued scalars, vectors, and matrices may appear a bit scary at first, you will soon notice that it is not really much more complicated than working with their real-valued counterparts.

**Homework 1.3.3.4.** Let $A \in \mathbb{C}^{m \times k}$ and $B \in \mathbb{C}^{k \times n}$. Using what you once learned about matrix transposition and matrix-matrix multiplication, reason that $(AB)^H = B^H A^H$.

▼ Solution

$(AB)^H$

$= \quad < X^H = \overline{X^T} >$

$\overline{(AB)^T}$

$= \quad <$ you once discovered that $(AB)^T = B^T A^T >$

$\overline{B^T A^T}$

$= \quad <$ you may check separately that $\overline{XY} = \overline{X}\,\overline{Y} >$

$\overline{B^T}\ \overline{A^T}$

$= \quad < \overline{X^T} = \overline{X}^T >$

$B^H A^H$

**Definition 1.3.3.4. Hermitian.** A matrix $A \in \mathbb{C}^{m \times m}$ is ***Hermitian*** if and only if $A = A^H$.

Obviously, if $A \in \mathbb{R}^{m \times m}$, then $A$ is a Hermitian matrix if and only if $A$ is a symmetric matrix.

**Homework 1.3.3.5.** Let $A \in \mathbb{C}^{m \times n}$.

ALWAYS/SOMETIMES/NEVER: $\| A^H \|_F = \| A \|_F$.

▶ Answer ▼ Solution

$\| A \|_F$

$= \quad <$ definition $>$

$\sqrt{\sum_{i=0}^{m-1} \sum_{j=0}^{n-1} |\alpha_{i,j}|^2}$

$= \quad <$ commutativity~of~addition $>$

$\sqrt{\sum_{j=0}^{n-1} \sum_{i=0}^{m-1} |\alpha_{i,j}|^2}$

$= \quad <$ change of variables $>$

$\sqrt{\sum_{i=0}^{n-1} \sum_{j=0}^{m-1} |\alpha_{j,i}|^2}$

$= \quad <$ algebra $>$

$\sqrt{\sum_{i=0}^{n-1} \sum_{j=0}^{m-1} |\overline{\alpha_{j,i}}|^2}$

$= \quad <$ definition $>$

$\| A^H \|_F$

Similarly, other matrix norms can be created from vector norms by viewing the matrix as a vector. It turns out that, other than the Frobenius norm, these aren't particularly interesting in practice. An example can be found in Homework 1.6.1.6.

**Remark 1.3.3.5.** The Frobenius norm of a $m \times n$ matrix is easy to compute (requiring $O(mn)$ computations). The functions $f(A) = \|A\|_F$ and $f(A) = \|A\|_F^2$ are also differentiable. However, you'd be hard-pressed to find a meaningful way of linking the definition of the Frobenius norm to a measure of an underlying linear transformation (other than by first transforming that linear transformation into a matrix).